Minseong Cho

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Softlab, Ajou University
Suwon, South Korea
Portfolio
Google Scholar

Research Interests

Active matter dynamics · Elastic Leidenfrost effect · Mechanics of moisture-driven hair frizz

Personal Statement

Currently a graduate researcher at Ajou University. My research interests include active matter dynamics, the Elastic Leidenfrost effect, and the mechanics of moisture-driven hair frizz.

Education

- **M.S. in Mechanical Engineering** Sep 2025 – Present
  Ajou University, Suwon, South Korea
  - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025
  Ajou University, Suwon, South Korea

Research Experience

- **Graduate Research Student** – Softlab, Ajou University Sep 2025 – Present
  - **Mechanics of moisture-driven hair frizz**

    Focus: moisture-driven changes in strand morphology.
    Techniques: centerline-based geometric analysis and simulation-assisted interpretation.
    Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung

  - **Boundary-driven active matter in soft hydrogels**

    Focus: Leidenfrost-induced dynamics in soft hydrogel systems.
    Techniques: image-based tracking, curvature analysis, and visualization scripts.
    Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung

- **Undergraduate Research Student** – Softlab, Ajou University Mar 2023 – Aug 2025
  - **Boundary-driven active matter in soft hydrogels**

    Focus: Leidenfrost-related soft-matter dynamics.
    Techniques: image-based tracking, quantitative analysis, and figure preparation.
    Advisor: Prof. Jonghyun Ha, Dr. Yeonsu Jung

Conferences and Presentations

- **APS Division of Fluid Dynamics Meeting (Houston, TX, USA)** Nov 2025
  - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **Asian Congress of Fluid Mechanics (Seoul, South Korea)** Sep 2025
  - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **KSME Annual Conference (Jeju, South Korea)** Nov 2024
  - Explosive motion of spherical hydrogel via the Leidenfrost effect

Awards & Honors

- **Dean's List** – Ajou University Spring 2024, Spring 2025
  - Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** – Ajou University June 2025
  2nd Place · KRW 1,000,000 prize
  - **Project:** Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025
  3rd Place · KRW 300,000 prize
  - **Organized by:** Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)
  - **Project:** Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** 2025
  US$750 travel cost reimbursement
  - **Organized by:** APS Division of Fluid Dynamics, American Physical Society
  - **Purpose:** Travel support for participation in the APS Division of Fluid Dynamics Meeting.
- **Gallery of Fluid Motion** Apr 2026
  Haecheon Choi GFM Award · KRW 400,000 prize
  - **Organized by:** The Korean Society of Mechanical Engineers, Fluid Engineering Division
  - **Title:** Explosive motion of spherical hydrogel via the Leidenfrost effect
  - **Authors:** Minseong Cho, Yeonsu Jung, Jonghyun Ha

Publications

- **Boundary-driven active matter and stochastic rectification via Leidenfrost impulses**
  Minseong Cho, Yeonsu Jung, and Jonghyun Ha
  In preparation

References

Prof. Jonghyun Ha
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